Electricty-Visualisation Milestone 1

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1 Problematic

We will study how geo-political conflicts affect the supply and demand of electricity in the world. Our initial hypothesis is that as a conflict surges, especially in energy producing regions (middle east), countries whose energy supply depends on these regions are highly affected and this should be reflected both in how the country produces their energy in reaction and in turn in prices. Our goal is to better understand how global conflict impact the dynamics of energy production and prices and to showcase this through a well built data-story. As described below, we aim to create a dynamic website designed to appeal to the young and middle-aged population so that they better understand how conflicts affect their daily lives through electricity prices. To do so, we will study over time the methods and quantity of electricity produced around the world and combine this with day-ahead prices of electricity across markets. Our first scratch of the project will be a cinematic scrollytelling experience that begins with a high-contrast split-screen portal, suggesting the user to choose between the Ukraine or Iran conflict as the primary catalyst for their journey. Once a path is selected, the website transforms its entire visual identity using cold blues for the European crisis or warm oranges for Middle Eastern shocks and guides the user through a narrative where data shapes literally shift and break as the user scrolls. You will move from a stable energy situation into a shock phase, using fluid flow maps and reactive line charts to visually demonstrate how distant geopolitical fractures ripple through the European energy grid and settle into the daily cost of living (pending data availability). This interactive investigation turns dry market statistics into a documentary that reveals the interconnectedness of global energy production and price volatility.

2 Dataset

To better understand the effect of geo-political conflicts on energy flows, we build a data-driven narrative exploring the relationship between energy production and electricity prices. To achieve this, it is essential to rely on high-quality

datasets for both production and pricing. We identified two datasets that meet these requirements:

1. Electricity production dataset from ElectricityMaps [3]: Using the ElectricityMaps API, we retrieve electricity production data for a large set of countries worldwide starting from 2021. The dataset includes multiple production sources, such as gas, coal, and hydroelectric power. The data quality is generally high globally and particularly strong within Europe. However, the format is specific to ElectricityMaps, requiring preprocessing and transformation into a suitable format. Additionally, the API returns many features, some of which are not relevant for our study, so we retain only the most useful variables.

2. Electricity price dataset from the ENTSO-E Transparency Platform [4]: We obtained electricity price data using the ENTSO-E API, covering multiple European countries from 2018 to 2025. The main preprocessing step consisted of merging country-level data into a unified dataset. The data is highly reliable: prices originate from power exchanges (e.g., EPEX SPOT, Nord Pool), which are required under EU regulations to report to Transmission System Operators (TSOs), who then publish them on the ENTSO-E platform. This ensures high accuracy and consistency.

Because such high-quality data is primarily available in Europe due to regulatory requirements, we restrict our analysis to the European continent. This allows us to prioritize data reliability and completeness over broader coverage.

Overall, data collection was straightforward, as we identified well-designed APIs providing free and structured access to high-quality data for academic use.

3 Exploratory Data Analysis

Exploratory Analysis of Electricity Production and Prices: We begin our analysis with the electricity production dataset. Across Europe, the energy mix is dominated by coal, gas, and hydroelectric sources, while geothermal energy represents a negligible share of total production. At the global level, total electricity generation amounts to approximately 142,557 TWh in 2021, with China emerging as the largest producer, contributing around 36,717 TWh.

To assess the stability of the system, we compute the coefficient of variation of total electricity production over time. We find that aggregate production fluctuates by approximately 6% around its mean level. This indicates that, despite the large scale of the system, electricity generation remains relatively stable, though it still exhibits economically meaningful variability. Such fluctuations are likely driven by a combination of seasonal demand patterns, renewable intermittency, and structural differences across countries but also geo-political conflicts. We will understand this later on in the project.

We next turn to electricity prices using ENTSO-E data over the period 2018–2026. The European energy market displays an average price level of 87.35

EUR/MWh, accompanied by a high standard deviation of 76.12 EUR/MWh. This substantial dispersion highlights the presence of significant price volatility, suggesting that the market is frequently subject to strong external shocks.

Importantly, these price fluctuations are not purely stochastic. Instead, they coincide with major geopolitical and economic disruptions. In particular, the sharp increase in prices observed in 2022 aligns with the onset of the Russia–Ukraine conflict, which severely constrained energy supply in Europe. As illustrated in Figure 1, these extreme price movements reflect periods during which the market is pushed far from equilibrium, leading to substantial increases in energy costs.

4 Related work

In [1] they studies how geopolitical events affect inflation in European countries through changes in energy prices. It shows that energy is a key driver of inflation when supply is disrupted but that the effect is not an important across all countries, and that the size of the impact is related to how the country produces its electricity.

Another resource is [2] where they study how geopolitical risks affect energy markets, particularly oil and gas prices. It shows that increases in geopolitical uncertainty tend to drive higher energy prices and volatility, as markets anticipate supply disruptions and risk premiums rise. The impact is especially strong during major global events and differs across regions depending on their energy dependence and exposure to risk.

Finally in the form of a website, the IEA has studied the 2022 Energy Crisis in [5]. Although the explanations are well detailed and helps us understand the very specific case of the crisis of 2022 they do not showcase data in a visualisation manner.

Overall after a thorough study of related works we have not found a data-story explaining visually how geopolitical situations and energy markets are related. With this in mind we believe that our visual approach will be original compared to what is currently out.

References

[1] Oladiran B. Adekoya, John A. Oliyide, et al. "Geopolitical turmoil and energy dynamics: Analyzing the impact on inflation in selected European economies". In: *Heliyon* 11.3 (2025), e25682. DOI: 10.1016/j.heliyon.2025.e25682. URL: https://www.sciencedirect.com/science/article/pii/S2405844025006826.

[2] Dora Almeida et al. "Exploring the connection between geopolitical risks and energy markets". In: *Energy Economics* 141 (2025), p. 108113. ISSN: 0140-9883. DOI: https://doi.org/10.1016/j.eneco.2024.108113. URL: https://www.sciencedirect.com/science/article/pii/S0140988324008223.

[3] Electricity Maps. *Electricity Maps API*. `https://api.electricitymaps.com/`. Accessed: 2026-03-17. 2024.

[4] ENTSO-E. *ENTSO-E Transparency Platform*. `https://transparency.entsoe.eu/`. Accessed: 2026-03-17. 2024.

[5] International Energy Agency. *2022 Energy Crisis*. `https://www.iea.org/topics/2022-energy-crisis`. Accessed: 2026-03-17. 2022.

5 Appendix

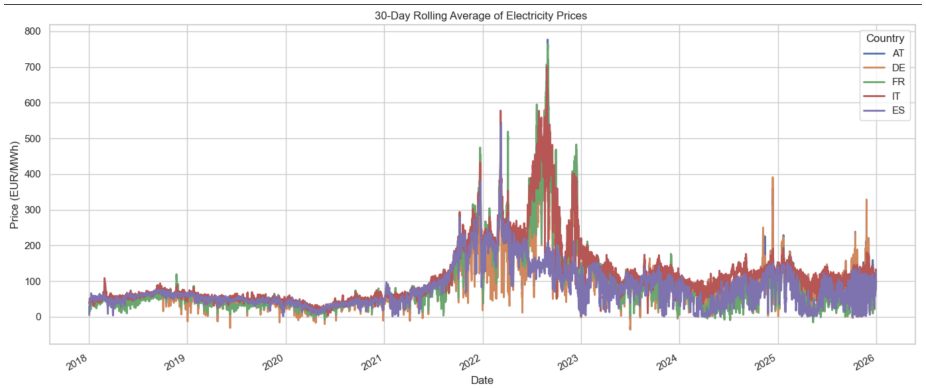

Figure 1: Prices overtime